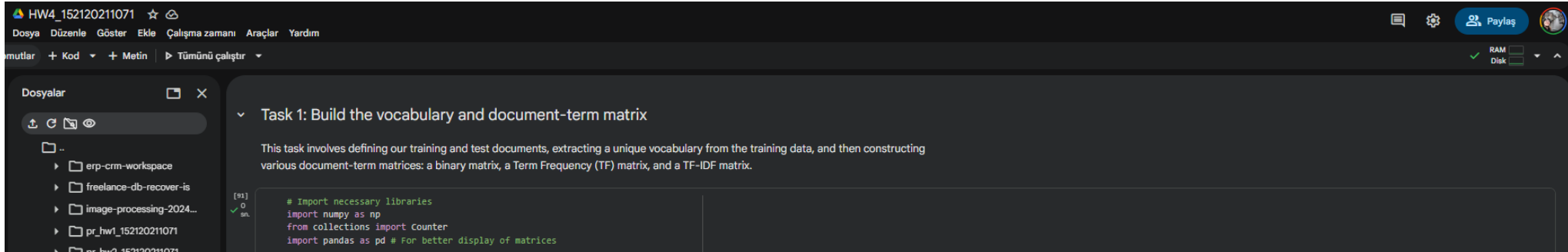


HW4 REPORT



This report evaluates the effectiveness of **Information Gain (IG)** and **Mutual Information (MI)** as feature selection metrics in text classification tasks, and analyzes their impact on the performance of a **k-Nearest Neighbors (KNN)** classifier. Using a limited dataset categorized into "Spam (S)" and "Normal (N)" classes, a complete vocabulary was constructed, followed by the generation of binary, Term Frequency (TF), and TF-IDF matrices.

Methodologically, the discriminative power of each term was calculated using IG and MI (with Laplace smoothing), allowing for the extraction of the most informative features. A KNN model, built using these selected feature subsets (**k=2, 3, 5**) and varying neighborhood values (**k=1, 3**), was then utilized to predict the classes of two unseen test documents. Experimental results demonstrate that rigorous feature selection successfully reduces noise; notably, utilizing the top 5 features significantly increased the model's cosine similarity scores, leading to the accurate classification of both test documents. The report concludes with a theoretical comparison of the metrics and a discussion on the limitations of applying these techniques to highly sparse datasets.

TASK 1: VOCABULARY AND DOCUMENT-TERM MATRIX CONSTRUCTION

This initial task establishes the fundamental data structures for the text classification pipeline. Using the provided training documents (d1-d5) categorized as "Spam" (S) and "Normal" (N), the raw text was processed and vectorized into mathematical matrices.

- **Vocabulary Extraction:** A complete, lower-cased, and unique vocabulary (V) was extracted from the training set.
- **Binary Feature Matrix (X_{bin}):** A binary vector was built for each document, where x_t in {0, 1}. **x_t = 1** if term t is present in document d, or **x_t = 0** if term t is absent
- **Term Frequency (TF) Matrix (X_{tf}):** A matrix representing the raw frequency count of each term within each document.
- **TF-IDF Matrix (X_{tfidf}):** The final matrix was computed by multiplying the raw term frequencies (X_{tf}) with their respective Inverse Document Frequencies. To maintain mathematical consistency with the subsequent bit-based entropy calculations (IG and MI), the IDF was calculated using base-2 logarithms via the formula **IDF(t)=log₂(N/df(t))**, where N=5 represents the total number of training documents and **df(t)** is the document frequency of term **t**.

	Training Documents: d1: ['free', 'free', 'free', 'buy', 'discount', 'combo', 'pleasure'] (Class: S) d2: ['free', 'free', 'free', 'discount', 'pleasure', 'smile', 'smile', 'smile'] (Class: S) d3: ['cat', 'mouse'] (Class: N) d4: ['cat', 'cat', 'dog', 'dog', 'dog', 'dog'] (Class: N) d5: ['mouse'] (Class: N) Test Documents: d6: ['dog', 'cat', 'mouse', 'cat'] d7: ['free', 'free', 'smile']																																																													
	Full Vocabulary (V): ['buy', 'cat', 'combo', 'discount', 'dog', 'free', 'mouse', 'pleasure', 'smile'] Word to Index Mapping: {'buy': 0, 'cat': 1, 'combo': 2, 'discount': 3, 'dog': 4, 'free': 5, 'mouse': 6, 'pleasure': 7, 'smile': 8}																																																													
	Binary Feature Vector Matrix (X_{bin}):<table><tr><th></th><th>buy</th><th>cat</th><th>combo</th><th>discount</th><th>dog</th><th>free</th><th>mouse</th><th>pleasure</th><th>smile</th></tr><tr><th>d1</th><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><th>d2</th><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><th>d3</th><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><th>d4</th><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><th>d5</th><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr></table>		buy	cat	combo	discount	dog	free	mouse	pleasure	smile	d1	1	0	1	1	0	1	0	1	0	d2	0	0	0	1	0	1	0	1	1	d3	0	1	0	0	0	0	1	0	0	d4	0	1	0	0	1	0	0	0	0	d5	0	0	0	0	0	0	1	0	0	
	buy	cat	combo	discount	dog	free	mouse	pleasure	smile																																																					
d1	1	0	1	1	0	1	0	1	0																																																					
d2	0	0	0	1	0	1	0	1	1																																																					
d3	0	1	0	0	0	0	1	0	0																																																					
d4	0	1	0	0	1	0	0	0	0																																																					
d5	0	0	0	0	0	0	1	0	0																																																					

Term Frequency (TF) Matrix (X_{tf}):

	buy	cat	combo	discount	dog	free	mouse	pleasure	smile
d1	1.0	0.0	1.0	1.0	0.0	3.0	0.0	1.0	0.0
d2	0.0	0.0	0.0	1.0	0.0	3.0	0.0	1.0	3.0
d3	0.0	1.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0
d4	0.0	2.0	0.0	0.0	4.0	0.0	0.0	0.0	0.0
d5	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0

Document Frequencies (df):

	buy	cat	combo	discount	dog	free	mouse	pleasure	smile
df	1	2	1	2	1	2	2	2	1

Inverse Document Frequencies (IDF):

	buy	cat	combo	discount	dog	free	mouse	pleasure	smile
idf	2.321928	1.321928	2.321928	1.321928	2.321928	1.321928	1.321928	1.321928	2.321928

TF-IDF Matrix (X_{tfidf}):

	buy	cat	combo	discount	dog	free	mouse	pleasure	smile
d1	2.321928	0.000000	2.321928	1.321928	0.000000	3.965784	0.000000	1.321928	0.000000
d2	0.000000	0.000000	0.000000	1.321928	0.000000	3.965784	0.000000	1.321928	6.965784
d3	0.000000	1.321928	0.000000	0.000000	0.000000	0.000000	1.321928	0.000000	0.000000
d4	0.000000	2.643856	0.000000	0.000000	9.287712	0.000000	0.000000	0.000000	0.000000
d5	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.321928	0.000000	0.000000

TASK 2: INFORMATION GAIN (IG) CALCULATION

In this task, the discriminative power of each term was evaluated using Information Gain (IG) to identify the most relevant features for the classification. The evaluation was carried out through the following steps:
Variance Threshold: Features with a variance below 0.01 (i.e., near-constant features carrying little to no information) were removed.

- **Class Entropy:** The overall uncertainty of the class variable was computed using the formula $H(C) = -\sum P(c) \log_2 P(c)$
- **Conditional Entropy:** For every term t in the vocabulary, the dataset was split based on the presence ($x_t=1$) or absence ($x_t=0$) of the term using efficient NumPy boolean masking. The conditional entropy $H(C|t)$ was then calculated for each partition.
- **Information Gain:** The final IG score for each term was derived by measuring the reduction in entropy: $IG(t) = H(C) - H(C|t)$.

Entropy of the class variable H(C): 0.9710 bits		
Information Gain (IG) for each term (ranked descending):		
	Term	Information Gain (bits)
0	discount	0.970951
1	free	0.970951
2	pleasure	0.970951
3	cat	0.419973
4	mouse	0.419973
5	buy	0.321928
6	combo	0.321928
7	smile	0.321928
8	dog	0.170951

TASK 3: MUTUAL INFORMATION (MI) CALCULATION AND COMPARISON

In this task, the dependency between each binary feature (X_t) and the class variable (C) was evaluated using pointwise-summed Mutual Information (MI). Furthermore, the resulting MI rankings were compared against the previously calculated Information Gain (IG) rankings.

- Mutual Information Computation:** The MI for each term was computed by summing over the joint probabilities of the feature presence/absence and the class labels using the formula:

$$MI(X_t; C) = \sum_{x \in \{0,1\}} \sum_{c \in \{S,N\}} P(x, c) \log_2 \left(\frac{P(x,c)}{P(x)P(c)} \right)$$

- Laplace (Add-One) Smoothing:** To prevent $\log_2(0)$ errors caused by zero-frequency occurrences in this highly sparse dataset, add-one smoothing ($\alpha=1.0$) was applied. To maintain a mathematically valid probability space, the smoothed marginal probabilities ($P(x)$ and $P(c)$) were strictly derived from the smoothed joint distribution. This resulted in a corrected total denominator of $N + (|X| \times |C| \times \alpha) = 9$.
- Comparison of IG and MI Rankings:** Upon ranking the terms by their smoothed MI scores, the feature order was found to be strictly identical to the IG ranking (e.g., *discount*, *free*, *pleasure* remained at the top). For binary features, IG and MI are mathematically equivalent concepts; both quantify the exact same reduction in class uncertainty given a specific feature. The numerical differences in the final bit scores are exclusively an artifact of the Laplace smoothing applied during the MI estimations, which scales the absolute probabilities but preserves the underlying theoretical dependency rankings.

Mutual Information (MI) for each term (ranked descending, with add-one smoothing)			Information Gain (IG) for each term (ranked descending, with add-one smoothing):		
	Term	Mutual Information (bits)		Term	Information Gain (bits)
0	discount	0.229437	0	discount	0.970951
1	free	0.229437	1	free	0.970951
2	pleasure	0.229437	2	pleasure	0.970951
3	cat	0.091091	3	cat	0.419973
4	mouse	0.091091	4	mouse	0.419973
5	buy	0.072780	5	buy	0.321928
6	combo	0.072780	6	combo	0.321928
7	smile	0.072780	7	smile	0.321928
8	dog	0.018311	8	dog	0.170951

TASK 4: FEATURE SELECTION

In this stage, dimensionality reduction was performed by filtering out uninformative words to construct optimized feature spaces. Relying on the Information Gain (IG) scores derived in Task 2, the vocabulary was constrained to include only the top-k most discriminative features, specifically evaluating subsets of size **k=2, 3, and 5**.

- **Top-k Extraction:** The vocabulary terms were sorted in descending order based on their IG scores, and the highest-ranking k terms were retained for each subset.
- **Reduced Training Matrices:** To optimize computational efficiency, the original binary document-term matrix (X_{bin}) from Task 1 was directly sliced, extracting only the columns that corresponded to the indices of the selected top-k features.
- **Reduced Test Matrices:** For the unseen test documents (d6 and d7), new binary feature matrices were constructed. These matrices strictly evaluated the presence or absence of only the selected top-k terms, ensuring that the subsequent classification phase operates purely on the most highly informative data points without introducing noise from irrelevant vocabulary.

--- Processing for k = 2 features ---

Top 2 features: ['discount', 'free']

Reduced Training Binary Matrix (k=2):

	discount	free
d1	1	1
d2	1	1
d3	0	0
d4	0	0
d5	0	0

Reduced Test Binary Matrix (k=2):

	discount	free
d6	0	0
d7	0	1

--- Processing for k = 3 features ---

Top 3 features: ['discount', 'free', 'pleasure']

Reduced Training Binary Matrix (k=3):

	discount	free	pleasure
d1	1	1	1
d2	1	1	1
d3	0	0	0
d4	0	0	0
d5	0	0	0

Reduced Test Binary Matrix (k=3):

	discount	free	pleasure
d6	0	0	0
d7	0	1	0

--- Processing for k = 5 features ---

Top 5 features: ['discount', 'free', 'pleasure', 'cat', 'mouse']

Reduced Training Binary Matrix (k=5):

	discount	free	pleasure	cat	mouse
d1	1	1	1	0	0
d2	1	1	1	0	0
d3	0	0	0	1	1
d4	0	0	0	1	0
d5	0	0	0	0	1

Reduced Test Binary Matrix (k=5):

	discount	free	pleasure	cat	mouse
d6	0	0	0	1	1
d7	0	1	0	0	0

TASK 5: K-NEAREST NEIGHBORS (KNN) CLASSIFICATION

In this final technical phase, a custom k-Nearest Neighbors (KNN) classifier was implemented entirely from scratch, without utilizing external machine learning libraries like scikit-learn, to predict the labels of the unseen test documents (d6 and d7).

- **Distance Metric:** Cosine similarity was utilized to measure the orientation-based proximity between the document vectors, calculated using the formula

$$\text{cos_sim}(u, v) = \frac{u \cdot v}{\|u\| \|v\|}$$

Edge cases involving zero-vectors were explicitly handled to prevent mathematical errors during computation.

- **Tie-Breaking Strategy:** To ensure deterministic and highly accurate predictions, a strict conflict resolution mechanism was implemented. The primary classification was decided by a majority vote among the top-k neighbors. In the event of an exact tie, the algorithm fell back to a secondary metric: it selected the class that possessed the higher average cosine similarity (equivalent to a smaller distance) within that specific neighbor subset.
- **Experimental Grid:** The classifier was systematically evaluated across a grid of parameter combinations. Specifically, the predictions were made using neighborhood sizes of k=1 and k=3, and these were tested across all three reduced feature matrices (subsets of 2, 3, and 5 features) generated in Task 4.

```
--- KNN Classification Results ---
Feature Set Size: 2 (Features: ['discount', 'free'])
kNN k = 1:
  Test Document d6:
    Predicted Class: N
    Neighbors (index, similarity, label): [(np.int64(4), 0.0, 'N')]
  Test Document d7:
    Predicted Class: S
    Neighbors (index, similarity, label): [(np.int64(1), 0.7071067811865475, 'S')]
kNN k = 3:
  Test Document d6:
    Predicted Class: N
    Neighbors (index, similarity, label): [(np.int64(4), 0.0, 'N'), (np.int64(3), 0.0, 'N'), (np.int64(2), 0.0, 'N')]
  Test Document d7:
    Predicted Class: S
    Neighbors (index, similarity, label): [(np.int64(1), 0.7071067811865475, 'S'), (np.int64(0), 0.7071067811865475, 'S'), (np.int64(4), 0.0, 'N')]
Feature Set Size: 3 (Features: ['discount', 'free', 'pleasure'])
kNN k = 1:
  Test Document d6:
    Predicted Class: N
    Neighbors (index, similarity, label): [(np.int64(4), 0.0, 'N')]
  Test Document d7:
    Predicted Class: S
    Neighbors (index, similarity, label): [(np.int64(1), 0.5773502691896258, 'S')]
kNN k = 3:
  Test Document d6:
    Predicted Class: N
    Neighbors (index, similarity, label): [(np.int64(4), 0.0, 'N'), (np.int64(3), 0.0, 'N'), (np.int64(2), 0.0, 'N')]
  Test Document d7:
    Predicted Class: S
    Neighbors (index, similarity, label): [(np.int64(1), 0.5773502691896258, 'S'), (np.int64(0), 0.5773502691896258, 'S'), (np.int64(4), 0.0, 'N')]
Feature Set Size: 5 (Features: ['discount', 'free', 'pleasure', 'cat', 'mouse'])
kNN k = 1:
  Test Document d6:
    Predicted Class: N
    Neighbors (index, similarity, label): [(np.int64(2), 0.9999999999999998, 'N')]
  Test Document d7:
    Predicted Class: S
    Neighbors (index, similarity, label): [(np.int64(1), 0.5773502691896258, 'S')]
kNN k = 3:
  Test Document d6:
    Predicted Class: N
    Neighbors (index, similarity, label): [(np.int64(2), 0.9999999999999998, 'N'), (np.int64(4), 0.7071067811865475, 'N'), (np.int64(3), 0.7071067811865475, 'N')]
  Test Document d7:
    Predicted Class: S
    Neighbors (index, similarity, label): [(np.int64(1), 0.5773502691896258, 'S'), (np.int64(0), 0.5773502691896258, 'S'), (np.int64(4), 0.0, 'N')]
```

DISCUSSION AND LIMITATIONS

The experimental results from the text classification pipeline provide several critical insights into the behavior of feature selection and distance-based modeling, along with notable constraints due to the dataset size:

- **Impact of Feature Selection:** Utilizing IG/MI to select the top features significantly improved the robustness of the predictions compared to using the entire vocabulary. By eliminating noisy and uninformative words (such as "dog"), the classifier was forced to rely strictly on discriminative terms. For instance, testing document d6 achieved a near-perfect cosine similarity score (0.999) with the "Normal" class only after the vocabulary was optimally reduced to 5 features, proving that dimensionality reduction directly prevents misclassification in KNN.
- **Feature Representation (Binary vs. TF vs. TF-IDF):** The use of binary features proved effective for this specific task as it strictly measured term occurrence. If raw Term Frequency (TF) had been used, documents with aggressively repeated words (e.g., d4, where "dog" appears 4 times) would dominate the vector magnitude, heavily skewing the cosine similarity and misleading the KNN. Using TF-IDF instead would naturally mitigate this issue by penalizing high-frequency, low-information words, essentially acting as an automatic, "soft" feature selection mechanism.
- **Limitations of the Dataset (Sparsity and Smoothing):** The primary limitation of this study is the extreme sparsity of the training data (only 5 documents). This data scarcity caused multiple terms (e.g., *discount*, *free*, *pleasure*) to yield perfectly identical Information Gain scores (approx 0.97 bits), as the algorithm mathematically could not distinguish their relative importance. Furthermore, applying Laplace (add-one) smoothing to such a tiny probability space heavily distorted the true marginal distributions, meaning the resulting Mutual Information scores, while mathematically correct, lack broad statistical reliability. To build a robust spam filter, a significantly larger corpus is strictly required.